

# ME315\_Lab02\_280867

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```
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

`filter`, `lag`

Os seguintes objetos são mascarados por 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
library(readr)
library(ggplot2)
library(ggcal)
```

## 1. Estatísticas Suficientes:

`n_atraso` = total de atrasos (`ARRIVAL_DELAY > 10`)

`n_total` = total de voos válidos

Essas duas estatísticas são o suficiente para calcular a razão de voos atrasados por dia por cia aérea.

## 2. Função getStats

```

getStats <- function(input, pos) {
  input %>%
    filter(AIRLINE %in% c("AA", "DL", "UA", "US"),
           !is.na(ARRIVAL_DELAY), !is.na(MONTH), !is.na(DAY)) %>%
    group_by(AIRLINE, MONTH, DAY) %>%
    summarise(
      n_atraso = sum(ARRIVAL_DELAY > 10),
      n_total  = n(),
      .groups = "drop"
    )
}

```

3. Importação do arquivo `flights.csv.zip` e configuração de colunas

```

mycols <- cols_only(
  MONTH      = 'i',
  DAY        = 'i',
  AIRLINE    = 'c',
  ARRIVAL_DELAY = 'd'
)

stats_por_lote <- read_csv_chunked(
  file = "flights.csv.zip",
  callback = DataFrameCallback$new(getStats),
  chunk_size = 100000,
  col_types = mycols
)

```

4. `computeStats`

```

computeStats <- function(x) {
  x %>%
    group_by(AIRLINE, MONTH, DAY) %>%
    summarise(
      n_atraso = sum(n_atraso),
      n_total  = sum(n_total),
      .groups = "drop"
    ) %>%
    mutate(
      Cia = AIRLINE,
      Data = as.Date(paste(2015, MONTH, DAY, sep = "-")),
      Perc = n_atraso / n_total
    )
}

```

```

    ) %>%
    select(Cia, Data, Perc)
}

stats <- computeStats(stats_por_lote)

```

## 5. Mapa de calor como calendário

```

pal <- scale_fill_gradient(low = "#4575b4", high = "#d73027")

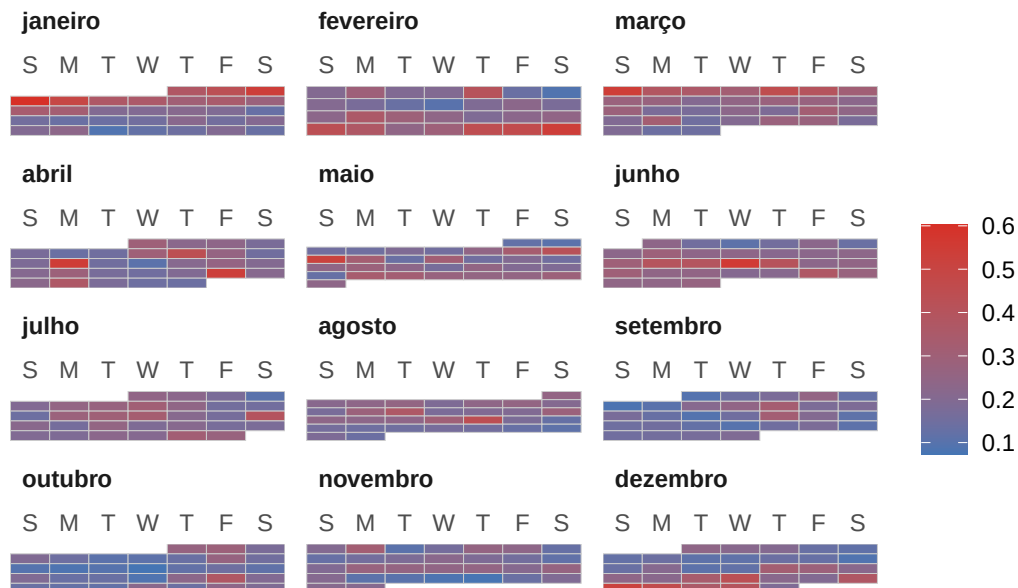
baseCalendario <- function(stats, cia) {
  sub <- stats %>% filter(Cia == cia)
  ggcal(sub$Data, sub$Perc)
}

cAA <- baseCalendario(stats, "AA")
cDL <- baseCalendario(stats, "DL")
cUA <- baseCalendario(stats, "UA")
cUS <- baseCalendario(stats, "US")

cAA + pal + ggtitle("Percentual de Atrasos - AA")

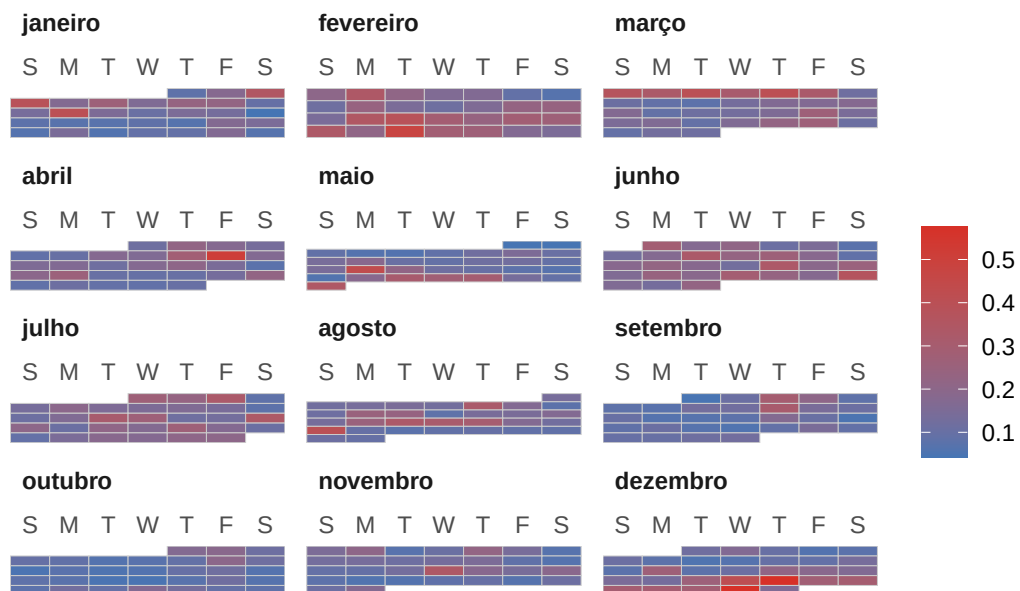
```

### Percentual de Atrasos – AA



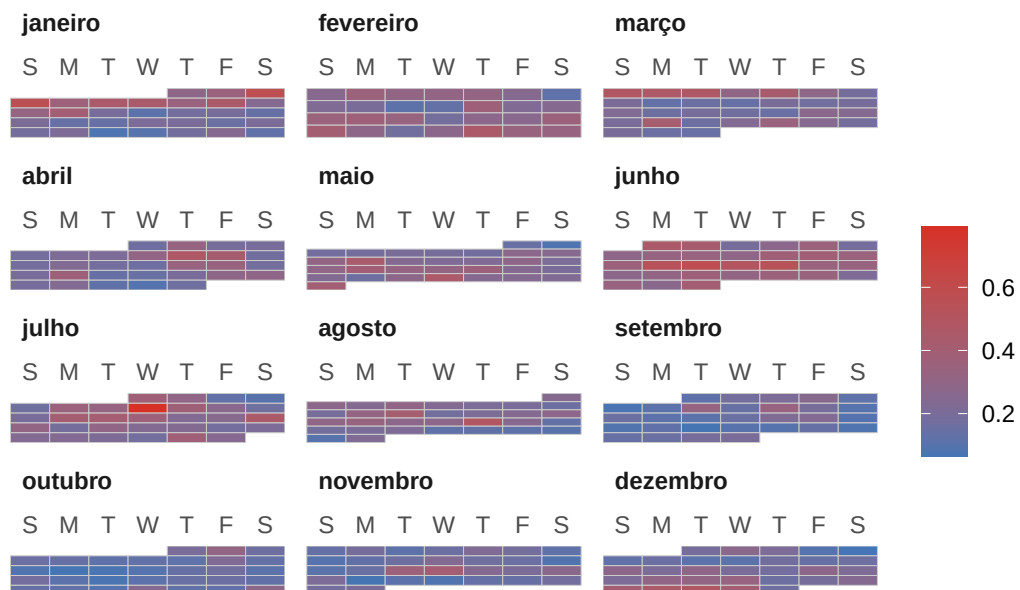
```
cDL + pal + ggtitle("Percentual de Atrasos - DL")
```

## Percentual de Atrasos – DL



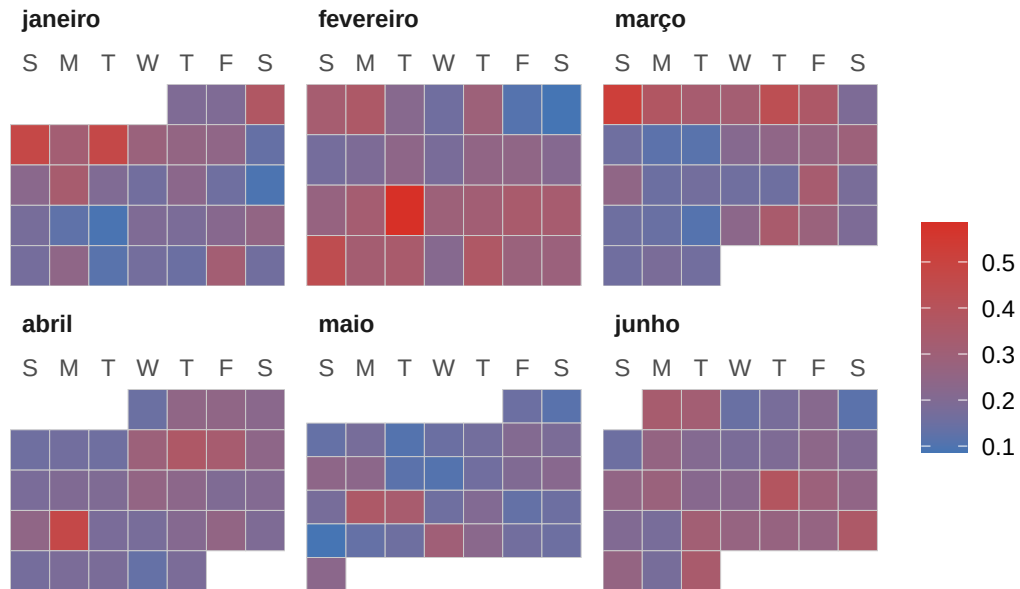
```
cUA + pal + ggtitle("Percentual de Atrasos - UA")
```

## Percentual de Atrasos – UA



```
cUS + pal + ggtitle("Percentual de Atrasos - US")
```

## Percentual de Atrasos – US



```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-24 11:25:23 -03"
```